

Project Design Phase

Solution Architecture

Date	27 September 2026
Team ID	_____
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Solution Architecture

The proposed solution uses ServiceNow Flow Designer to streamline the procurement of standard laptops. The architecture connects the employee request, approval process, procurement activities, and fulfillment workflow so that standard laptop orders can be processed consistently and with less manual effort.

Architecture and Data Flow

1. Employee Request	Employee submits a standard laptop request through the ServiceNow request interface.
2. Request Validation	The system checks the request details and confirms that the selected laptop is a standard item.
3. Approval	Flow Designer routes the request to the appropriate approver based on the defined process.
4. Procurement	After approval, the workflow creates or updates the required procurement activities.
5. Fulfillment	The approved laptop order is processed and assigned for fulfillment.
6. Status Update	Request status and relevant task information are updated automatically for tracking.

Key Solution Components

- **ServiceNow Request Interface:** Captures standard laptop requirements from the employee.
- **Flow Designer:** Automates routing, approvals, task creation, and workflow progression.
- **Approval Logic:** Ensures the request follows the required authorization process.
- **Procurement Tasks:** Supports consistent handoff from approval to purchasing and fulfillment.
- **Tracking:** Maintains request and task status so stakeholders can follow progress.

Expected Outcome

The architecture provides a structured workflow for standard laptop procurement, reduces repetitive manual steps, improves process consistency, and provides better visibility into request and fulfillment status.

Figure 1: Proposed architecture and data flow for automated standard laptop procurement

